Experiment 3

Aim

1. To design and simulate a 10 GHz microwave directional coupler and two-way power divider using MATLAB, and analyze coupling, isolation, insertion loss, directivity, and power division performance over the frequency range of 8–12 GHz.

Objectives

2. To model a 10 GHz directional coupler and two-way power divider using 50 Ω ports.
3. To calculate coupling factor, isolation, insertion loss, and directivity using S-parameters.
4. To analyze the equal power division ratio of a two-way power divider.
5. To compare the performance of the original coupler with a 5% increase in coupling-section length.
6. To evaluate the simulated results and identify the design with improved coupling, isolation, and insertion loss.

Software Required: MATLAB

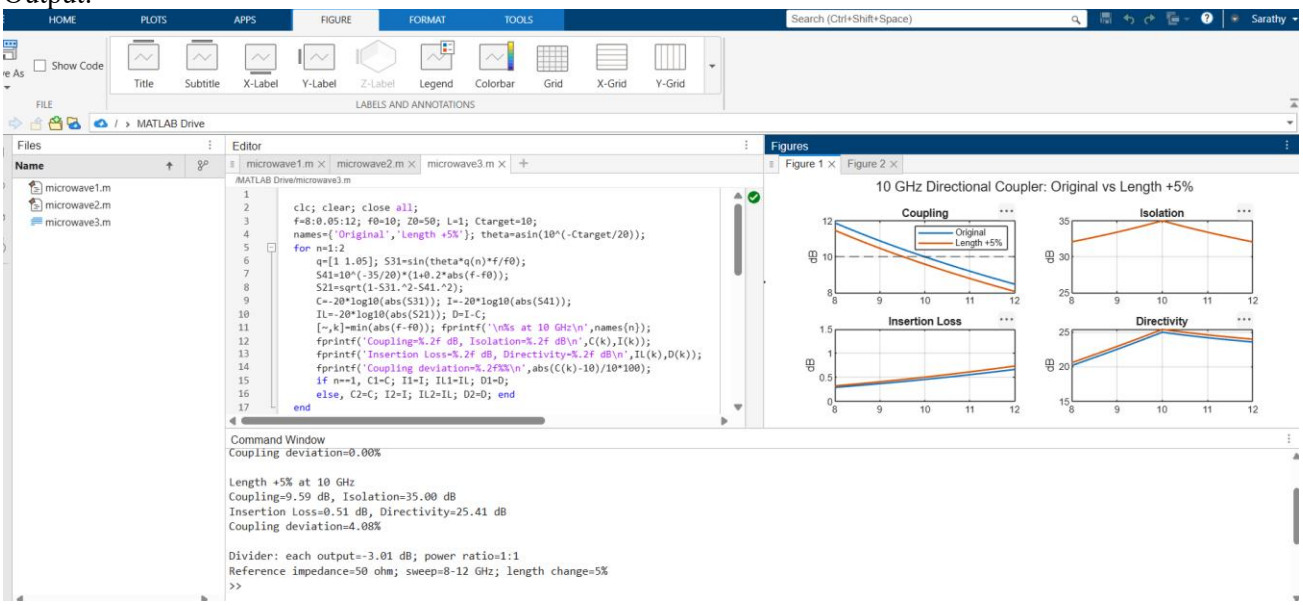
Code:

```
clc; clear; close all;
f=8:0.05:12; f0=10; Z0=50; L=1; Ctarget=10;
names={'Original','Length +5%'}; theta=asin(10^(-Ctarget/20));
for n=1:2
q=[1 1.05]; S31=sin(theta*q(n)*f/f0);
S41=10^(-35/20)*(1+0.2*abs(f-f0));
S21=sqrt(1-S31.^2-S41.^2);
C=-20*log10(abs(S31)); I=-20*log10(abs(S41));
IL=-20*log10(abs(S21)); D=I-C;
[~,k]=min(abs(f-f0)); fprintf('\n%s at 10 GHz\n',names{n});
fprintf('Coupling=%.2f dB, Isolation=%.2f dB\n',C(k),I(k));
fprintf('Insertion Loss=%.2f dB, Directivity=%.2f dB\n',IL(k),D(k));
fprintf('Coupling deviation=%.2f%%\n',abs(C(k)-10)/10*100);
if n==1, C1=C; I1=I; IL1=IL; D1=D;
else, C2=C; I2=I; IL2=IL; D2=D; end endfigure('Color','w');
subplot(2,2,1); plot(f,C1,f,C2,'LineWidth',1.4); yline(10,'--'); title('Coupling');
ylabel('dB'); grid on; legend(names);
subplot(2,2,2); plot(f,I1,f,I2,'LineWidth',1.4); yline(25,'--'); title('Isolation');
ylabel('dB'); grid on;
subplot(2,2,3); plot(f,IL1,f,IL2,'LineWidth',1.4); yline(1.5,'--'); title('Insertion
Loss'); ylabel('dB'); grid on;
subplot(2,2,4); plot(f,D1,f,D2,'LineWidth',1.4); yline(15,'--'); title('Directivity');
ylabel('dB'); grid on;
sgtitle('10 GHz Directional Coupler: Original vs Length +5%');
figure('Color','w'); divider=-3.0103+0.1*(f-f0).^2;
plot(f,divider,'b',f,divider,'r--','LineWidth',1.5); grid on;
title('Two-Way Equal Power Divider'); xlabel('Frequency (GHz)'); ylabel('Output
Transmission (dB)');
legend('Output Port 2','Output Port 3');
fprintf('\nDivider: each output=%.2f dB; power ratio=1:1\n',-3.0103);
fprintf('Reference impedance=%.0f ohm; sweep=8-12 GHz; length change=5%%\n',Z0);
```

➔ INPUT PARAMETER TABLE – VALUES TO BE ENTERED

S.No.	Input Parameter	Symbol / Setting	Value to Enter	Unit
1	Design frequency	f0	10	GHz
2	Reference impedance	Z0	50	Ω
3	Frequency sweep	fstart–fstop	8–12	GHz
4	Coupling target	C	10	dB
5	Port configuration	—	3-port two-way divider / 4-port coupler	—

Output:



Exp 5

a) Coupling factor

$C = -20 \log_{10} |S_{31}|$

For the target coupling of 10 dB

$10 = -20 \log_{10} |S_{31}|$

$|S_{31}| = 10^{-10/20}$

$|S_{31}| = 0.3162$

$\frac{P_3}{P_{in}} = |S_{31}|^2 = 0.10 = 10\%$

b) Isolation

$I = -20 \log_{10} |S_{41}|$

$|S_{41}| = 10^{-35/20}$

$|S_{41}| = 0.01778$

$\frac{P_4}{P_{in}} = |S_{41}|^2 = 0.0316\%$

c) Insertion loss

$IL = -20 \log_{10} |S_{21}|$

$IL = 0.46\,dB$

$|S_{21}| = 10^{-0.46/20}$

$|S_{21}| = 0.9484$

$\frac{P_2}{P_{in}} = |S_{21}|^2 = 89.94\%$

d) Directivity

$D = I - C = 35 - 10 = 25\,dB$

Length increased by 5%.

i) $C = 9.56\,dB$

$|S_{31}| = 10^{-9.56/20}$

$|S_{31}| = 0.3314$

ii) Coupled arm

$\% = \frac{|9.57 - (10)|}{10} \times 100$

$= 4.1\%$

iii) Isolation

$|S_{41}| = 10^{-35/20}$

$= 0.01778$

iv) Directivity

$D = I - C$

$D = 35 - 9.59$

$D = 25.41\,dB$

Two-way power divider

$S_{21} = S_{32} = -3.01\,dB$